

## Coding & Solution

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|----------------------|---------------|
| <b>Date</b>          | 30 June 2026  |
| <b>Team ID</b>       |               |
| <b>Project Name</b>  | Rising Waters |
| <b>Maximum Marks</b> | 5 Marks       |

## Coding & Solution

### Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

### Solution Summary

| Field                                | Details                                                                                                                              |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>Repository Link / URL</b>         | [Paste your GitHub repository link here]                                                                                             |
| <b>Programming Language(s)</b>       | Python, JavaScript, HTML, CSS                                                                                                        |
| <b>Framework(s) Used</b>             | Flask, Scikit-learn, XGBoost                                                                                                         |
| <b>Key Features Implemented</b>      | Flood prediction using machine learning models, real-time prediction through a Flask web application, risk classification dashboard. |
| <b>Pending / Incomplete Features</b> | Advanced alert notifications and live weather API integration.                                                                       |
| <b>Setup / Run Instructions</b>      | Install required libraries, run app.py, and access the Flask application through a web browser.                                      |

**Code Quality Checklist**

| S.No | Criteria                                               | Status (Yes / No) |
|------|--------------------------------------------------------|-------------------|
| 1    | Code is modular and organized into functions / classes | Yes               |
| 2    | Meaningful variable and function names are used        | Yes               |
| 3    | Code includes comments / documentation where necessary | Yes               |
| 4    | Error handling is implemented for critical operations  | Yes               |
| 5    | The application runs without critical errors           | Yes               |
| 6    | Code is committed to a version control repository      | Yes               |

**Additional Notes / Comments:**

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